

Pants on fire

Problem ID: pantsonfire

So it is so true Donald and Mike are the leaders of the free world and haven't yet (after half a year) managed to start a nuclear war. It is so great! It is so tremendous!

Despite the great and best success of Donald's Administration, there are still a few things he likes to complain about.

... some quote of DJT about Mexicans.

But he also frequently compares Mexicans to other people (like North-Koreans). Due to the tremendous amount of statements he has made (mostly containing less than 140 characters ...) the "Fake-News" New York Times has to put in a lot of effort to clarify and comment on all the statements of Donald. To check a statement, they have a list of facts they deem to be true and classify Donald's statements into three groups: real facts (which are logical conclusions from their list true facts), exaggerations (which do not follow, but are still consistent with the papers list of facts), and alternative facts (which contradict the knowledge of the newspaper).

They have asked you to write a program helping them to classify all of DJTs statements – after all it is hard for a journalist to go through them all and check them all, right?

Input

The input starts with a line containing two numbers N and M ($1 \leq N, M, \leq 200$), the number of facts deemed true by the NYT and the number of statements uttered by the Donald. Thereafter follow $N + M$ lines, each containing either

- S "are the worst", for some string S (no longer than 30 characters), stating that S is worse than all other peoples.
- A "are worse than" B , for some strings A and B (no longer than 30 characters), stating that A is (strictly) worse than B . A and B are never identical.

Note that Donald's statements may contain countries that the NYT does not know about (just like Jon Snow). We assume that worseness is transitive. You may assume that the first N lines do not contain any contradictory statement.

Interestingly, Donald's press secretary (Grumpy Sean) has managed to convince him not to make up countries when tweeting, thus the input mentions at most 193 different countries.

Output

For every of the M statements of DJT output one line containing

- "Fact" if the statement is true given the N facts of the NYT
- "Alternative Fact" if the inversion of the statement is true given the N facts of the NYT
- "Pants on Fire" if the statement does not follow, but neither does its negation.

Sample Input 1

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4 5
Mexicans are worse than Americans
Russians are worse than Mexicans
NorthKoreans are the worst
Canadians are worse than Americans
Russians are worse than Americans
Mexicans are the worst
NorthKoreans are worse than Mexicans
NorthKoreans are worse than Germans
Mexicans are worse than Canadians
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Sample Output 1

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Fact
Alternative Fact
Fact
Fact
Pants on Fire
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